

What is a large language model (LLM), and how does it work?

A large language model (LLM) is a type of [artificial intelligence \(AI\)](#) program that can recognize and generate text, among other tasks. LLMs are trained on [huge sets of data](#) — hence the name "large." LLMs are built on [machine learning](#): specifically, a type of [neural network](#) called a transformer model.

In simpler terms, an LLM is a computer program that has been fed enough examples to be able to recognize and interpret human language or other types of complex data. Many LLMs are trained on data that has been gathered from the Internet — thousands or millions of gigabytes' worth of text. Some LLMs continue to [crawl the web](#) for more content after they are initially trained. But the quality of the samples impacts how well LLMs will learn natural language, so an LLM's programmers may use a more curated data set, at least at first.

LLMs use a type of machine learning called [deep learning](#) in order to understand how characters, words, and sentences function together. Deep learning involves the probabilistic analysis of unstructured data, which eventually enables the deep learning model to recognize distinctions between pieces of content without human intervention.

LLMs are then further trained via tuning: they are fine-tuned or prompt-tuned to the particular task that the programmer wants them to do, such as interpreting questions and generating responses, or translating text from one language to another.

Machine learning and deep learning

At a basic level, LLMs are built on machine learning. Machine learning is a subset of AI, and it refers to the practice of feeding a program large amounts of data in order to train the program how to identify features of that data without human intervention.

LLMs use a type of machine learning called deep learning. Deep learning models can essentially train themselves to recognize distinctions without human intervention, although some human fine-tuning is typically necessary.

Deep learning uses probability in order to "learn." For instance, in the sentence "The quick brown fox jumped over the lazy dog," the letters "e" and "o" are the most common, appearing four times each. From this, a deep learning model could conclude (correctly) that these characters are among the most likely to appear in English-language text.

Realistically, a deep learning model cannot actually conclude anything from a single sentence. But after analyzing trillions of sentences, it could learn enough to predict how to logically finish an incomplete sentence, or even generate its own sentences.

LLM neural networks

In order to enable this type of deep learning, LLMs are built on neural networks. Just as the human brain is constructed of neurons that connect and send signals to each other, an artificial neural network (typically shortened to "neural network") is constructed of network nodes that connect with each other. They are composed of several "layers": an input layer, an output layer, and one or more layers in between. The layers only pass information to each other if their own outputs cross a certain threshold.

LLM transformer models

The specific kind of neural networks used for LLMs are called transformer models. Transformer models are able to learn context — especially important for human language, which is highly context-dependent. Transformer models use a mathematical technique called self-attention to detect subtle ways that elements in a sequence relate to each other. This makes them better at understanding context than other types of machine learning. It enables them to understand, for instance, how the end of a sentence connects to the beginning, and how the sentences in a paragraph relate to each other.

This enables LLMs to interpret human language, even when that language is vague or poorly defined, arranged in combinations they have not encountered before, or contextualized in new ways. On some level they "understand" semantics in that they can associate words and concepts by their meaning, having seen them grouped together in that way millions or billions of times.

What is the relevance for the business case?

AN LLM would have minimal relevance for this forecasting problem. The company needs numerical demand forecasting to avoid overstocking, but LLMs are designed for language processing, not analysing structured time-series data. The only use for LLMs in this scenario is that they're easy to prompt without machine-learning expertise. Another possible use case for LLMs could be as support tools for explaining results and improving user interaction, while the actual forecasting should use a predictive AI model like ARIMA, Prophet, or XGBoost, which are specifically designed for numerical predictions.